



Mrityunjay Shukla
Computer Science & Engineering
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Examination	University	Institute	Year	CPI/%
Post Graduation	IIT Bombay	IIT Bombay	2024-	8.83
Graduation (in ECE dept)	MIT, Manipal	Manipal, Karnataka	2020-24	8.23
Intermediate/+2	CBSE	DPS, Kalyanpur	2019-20	87
Matriculation	CBSE	DPS, Kalyanpur	2018	93

PUBLICATIONS

- **M.Shukla***, S. Roy, S. Saha, B. Panda. **"PRowhammer: Propagating bit-flips from CPU to GPU"** Under review at the International Symposium on Computer Architecture (ISCA '26).
- S. Ahamed*, **M.Shukla**, K. Singla, S. Saha. **"Everything Depends on Your Hammer: A Systematic Rowhammer Attack Exploration on SPHINCS+"** Accepted at the International Conference on Computer Aided Design (ICCAD '25)

* denotes primary author.

RESEARCH WORK

- **GPU scheduling and prefetching for Ray tracing workloads**
(Ongoing Research | Guide: **Prof. Biswabandan Panda**) (Nov'25-Present)
 - Developed a comprehensive understanding of GPU-based ray tracing by analyzing state-of-the-art scheduling and prefetching mechanisms.
 - Quantitatively evaluated the interplay between scheduling and prefetching techniques to determine whether their combined effects enhance or degrade overall performance.
- **Investigating and Understanding the Security of Heterogeneous Systems**
(PhD Seminar | Guide: **Prof. Biswabandan Panda** and **Prof. Sayandeep Saha**) (Jan'25-May'25)
 - Gained a strong understanding of CPU-GPU architectures and their data-flow interactions within heterogeneous computing systems.
 - Analyzed how Rowhammer exploits DRAM behavior and evaluated how similar attack principles could target CPU-GPU systems.
 - Identified how architectural coupling between CPUs and GPUs can expose new vulnerability surfaces, highlighting potential cross-component Rowhammer attack vectors.

INTERNSHIPS

- **Project Intern | Indian Institute of Technology, Kanpur** (Feb'24-June'24)
 - Worked on fine-tuning of LLMs for domain-specific applications.
 - Compared **QLORA**, **LORA**, and **RAG** methods for adapting **Llama-2** to cybersecurity tasks.
 - Built retrieval-augmented generation with vector databases to reduce model hallucinations.
 - Deployed Llama-2 model using **Pytorch**, **Hugging Face**, and **LangChain** for inference.
- **Project Intern | Indian Institute of Technology, Guwahati** (July'23-Jan'24)
 - Conducted comprehensive analysis of NEWS data related to military operations in China, creating a bespoke dataset with advanced NLP techniques.
 - Developed an efficient web application using **NextJs**, **FastAPI**, **Vercel**, and **Render** to provide intuitive insights for informed decision-making.

COURSE PROJECTS

- **Rowhammer attack on discrete GPUs**
(CS 490: R&D Project | **Prof. Biswabandan Panda** and **Prof. Sayandeep Saha**) (Aug'24-Dec'24)
 - Studied how **Rowhammer** attacks are mounted on **CPU DRAM**, including reverse-engineering physical-to-DRAM address mappings to build the foundational understanding needed to investigate similar attacks on **GPU VRAMs**.
 - Analyzed **NVIDIA GPU architecture (RTX 4090 and RTX 3050)** and attempted to reverse-engineer their physical-to-DRAM address mappings using the **PTX discard instruction** as a step toward assessing **Rowhammer** vulnerabilities on GPUs.
- **Performance Optimization**
(CS 683: Advanced Computer Architecture | **Prof. Biswabandan Panda**) (Aug'24-Dec'24)

- Optimized 2D-convolution operation. Achieved 4-5x speedup using **blocked matrix multiplication, SIMD & software prefetching**.
 - Implemented an **IP-stride** and **complex-stride prefetcher**.
- **Distributed key-value store**
(CS 744: Design and Engineering of Computing Systems | **Prof. Purushottam Kulkarni**) (Aug'24-Dec'24)
 - Designed and implemented a distributed key-value store with caching and evaluated the effectiveness of caching.
- **Brain tumor image segmentation using UNet++**
(CS 725: Foundations of Machine Learning | **Prof. Preethi Jyothi**) (Aug'24-Dec'24)
 - Used different augmentations, such as vertical and horizontal techniques, to improve brain tumor segmentation on the BRATS'2018 dataset.

TECHNICAL SKILLS

- **Programming & Scripting Languages:** C, C++, CUDA, Python, Bash
- **Development & Debugging Tools:** GDB, Valgrind, Git, Vim
- **Profiling Tools :** NSight Compute, NSight Systems , Perf , Intel VTune
- **Simulation Tools :** GPGPUSim, VulkanSim
- **Frameworks & Libraries:** ReactJS, NextJS, Flask, Pytorch Numpy, Pandas, Scrapy

TEACHING

- **Head TA | CS230/231: Digital Logic Design and Computer Architecture** (Jul'25-Nov'25)
- **Head TA | CS101: Computer Programming and Utilization** (Jan'25-May'25)
- **Head TA | CS230/231: Digital Logic Design and Computer Architecture** (Jul'24-Nov'24)

EXTRACURRICULAR ACTIVITIES

- Volunteered as a TA for a computer architecture workshop organized by INAE.